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**УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ**  
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**ITMO University**

**ПРАКТИЧЕСКАЯ РАБОТА 4**

**По дисциплине** Промышленная маршрутизация и управление трафиком

**Тема работы** Эмуляция распределенной корпоративной сети связи, настройка iBGP, организация L3VPN, VPLS

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## **ВВЕДЕНИЕ**

Цель работы – изучить протоколы BGP, MPLS и правила организации L3VPN и VPLS.

## 1 ЧАСТЬ 1

В первой части было необходимо: настроить iBGP RR Cluster, настроить VRF на 3 роутерах, настроить RD и RT на 3 роутерах, настроить IP адреса в VRF, проверить связность между VRF, настроить имена устройств, сменить логины и пароли.

На рисунках 1 и 2 представлен yaml-файл для первой части работы.

```
name: lab4

mgmt:
  network: mgmt
  ipv4-subnet: 172.20.20.0/24

topology:
  defaults:
    kind: vr-ros
    image: vrnetlab/mikrotik_routeros:6.47.9
  nodes:
    R01.HKI:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.10
    R01.SPB:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.11
    R01.LND:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.12
    R01.NY:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.13
    R01.LBN:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.14
    R01.SVL:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.15
    PC1:
      kind: linux
      image: alpine:latest
      cmd: sleep infinity
      network-mode: none
      exec:
        - ip addr add 192.168.10.253/24 dev eth1
        - ip route add default via 192.168.10.1
    PC2:
      kind: linux
      image: alpine:latest
      cmd: sleep infinity
```

Рисунок 1 – yaml

```

network-mode: none
exec:
- ip addr add 192.168.20.253/24 dev eth1
- ip route add default via 192.168.20.1
PC3:
kind: linux
image: alpine:latest
cmd: sleep infinity
network-mode: none
exec:
- ip addr add 192.168.30.253/24 dev eth1
- ip route add default via 192.168.30.1
links:
- endpoints: ["R01.HKI:eth1", "R01.LND:eth1"]
- endpoints: ["R01.HKI:eth2", "R01.LBN:eth1"]
- endpoints: ["R01.LND:eth2", "R01.LBN:eth2"]
- endpoints: ["R01.HKI:eth3", "R01.SPB:eth1"]
- endpoints: ["R01.LND:eth3", "R01.NY:eth1"]
- endpoints: ["R01.LBN:eth3", "R01.SVL:eth1"]
- endpoints: ["R01.SPB:eth2", "PC1:eth1"]
- endpoints: ["R01.NY:eth2", "PC2:eth1"]
- endpoints: ["R01.SVL:eth2", "PC3:eth1"]

```

Рисунок 2 – yaml

На рисунке 3 представлен конфиг для роутера HKI.

```

[admin@R01.HKI] > /interface bridge add name=lo0
[admin@R01.HKI] > /ip address
[admin@R01.HKI] /ip address> add address=10.0.0.11/32 interface=lo0
[admin@R01.HKI] /ip address> add address=10.11.21.1/30 interface=ether2
[admin@R01.HKI] /ip address> add address=10.11.31.1/30 interface=ether3
[admin@R01.HKI] /ip address> add address=10.11.10.1/30 interface=ether4
[admin@R01.HKI] /ip address> /routing ospf instance set default router-id=10.0.0.11
[admin@R01.HKI] /routing ospf network
[admin@R01.HKI] /routing ospf network> add network=10.0.0.11/32 area=backbone
[admin@R01.HKI] /routing ospf network> add network=10.11.21.0/30 area=backbone
[admin@R01.HKI] /routing ospf network> add network=10.11.31.0/30 area=backbone
[admin@R01.HKI] /routing ospf network> add network=10.11.10.0/30 area=backbone
[admin@R01.HKI] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.11 lsr-id=10.0.0.11
[admin@R01.HKI] /routing ospf network> /mpls ldp interface
[admin@R01.HKI] /mpls ldp interface> add interface=ether2
[admin@R01.HKI] /mpls ldp interface> add interface=ether3
[admin@R01.HKI] /mpls ldp interface> add interface=ether4
[admin@R01.HKI] /mpls ldp interface> /routing bgp instance set default as=65000 router-id=10.0.0.11 redistribute-connected=yes
[admin@R01.HKI] /mpls ldp interface> /routing bgp peer
[admin@R01.HKI] /routing bgp peer> add name=to_LND remote-address=10.0.0.21 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.HKI] /routing bgp peer> add name=to_LBN remote-address=10.0.0.31 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.HKI] /routing bgp peer> add name=to_SPB remote-address=10.0.0.10 remote-as=65000 update-source=lo0 route-reflect=yes address-families=vpn4
[admin@R01.HKI] /routing bgp peer>

```

Рисунок 3 – HKI

На рисунке 4 представлен конфиг для роутера SPB.

```

[admin@R01.SPB] > /interface bridge add name=lo0
[admin@R01.SPB] > /ip address
[admin@R01.SPB] /ip address> add address=10.0.0.10/32 interface=lo0
[admin@R01.SPB] /ip address> add address=10.11.10.2/30 interface=ether2
[admin@R01.SPB] /ip address> /routing ospf instance set default router-id=10.0.0.10
[admin@R01.SPB] /routing ospf network
[admin@R01.SPB] /routing ospf network> add network=10.0.0.10/32 area=backbone
[admin@R01.SPB] /routing ospf network> add network=10.11.10.0/30 area=backbone
[admin@R01.SPB] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.10 lsr-id=10.0.0.10
[admin@R01.SPB] /routing ospf network> /mpls ldp interface add interface=ether2
[admin@R01.SPB] /routing ospf network> /routing bgp instance set default as=65000 router-id=10.0.0.10
[admin@R01.SPB] /routing ospf network> /routing bgp peer add name=to_RR_HKI remote-address=10.0.0.11 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.SPB] /routing ospf network> /ip route vrf add routing-mark-VRF_DEVOPS interface=ether3 route-distinguisher=65000:1 import-route-targets=65000:1 export-route-targets=65000:1
[admin@R01.SPB] /routing ospf network> /ip address add address=192.168.10.1/24 interface=ether3
[admin@R01.SPB] /routing ospf network> /routing bgp instance vrf add routing-mark-VRF_DEVOPS instance=default redistribute-connected=yes
[admin@R01.SPB] /routing ospf network>

```

Рисунок 4 – SPB

На рисунке 5 представлен конфиг для роутера LND.

```
[admin@R01.LND] > /interface bridge add name=lo0
[admin@R01.LND] > /ip address
[admin@R01.LND] /ip address> add address=10.0.0.21/32 interface=lo0
[admin@R01.LND] /ip address> add address=10.11.21.2/30 interface=ether2
[admin@R01.LND] /ip address> add address=10.21.31.1/30 interface=ether3
[admin@R01.LND] /ip address> add address=10.21.20.1/30 interface=ether4
[admin@R01.LND] /ip address> /routing ospf instance set default router-id=10.0.0.21
[admin@R01.LND] /ip address> /routing ospf network
[admin@R01.LND] /routing ospf network> add network=10.0.0.21/32 area=backbone
[admin@R01.LND] /routing ospf network> add network=10.11.21.0/30 area=backbone
[admin@R01.LND] /routing ospf network> add network=10.21.31.0/30 area=backbone
[admin@R01.LND] /routing ospf network> add network=10.21.20.0/30 area=backbone
[admin@R01.LND] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.21 lsr-id=10.0.0.21
[admin@R01.LND] /routing ospf network> /mpls ldp interface
[admin@R01.LND] /mpls ldp interface> add interface=ether2
[admin@R01.LND] /mpls ldp interface> add interface=ether3
[admin@R01.LND] /mpls ldp interface> add interface=ether4
[admin@R01.LND] /mpls ldp interface> /routing bgp instance set default as=65000 router-id=10.0.0.21 redistribute-connected=yes
[admin@R01.LND] /mpls ldp interface> /routing bgp peer
[admin@R01.LND] /routing bgp peer> add name=to_HKI remote-address=10.0.0.11 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.LND] /routing bgp peer> add name=to_LBN remote-address=10.0.0.31 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.LND] /routing bgp peer> add name=to_NYD remote-address=10.0.0.20 remote-as=65000 update-source=lo0 route-reflect=yes address-families=vpn4
[admin@R01.LND] /routing bgp peer> █
```

Рисунок 5 – LND

На рисунке 6 представлен конфиг для роутера NY.

```
[admin@R01.NY] > /interface bridge add name=lo0
[admin@R01.NY] > /ip address
[admin@R01.NY] /ip address> add address=10.0.0.20/32 interface=lo0
[admin@R01.NY] /ip address> add address=10.21.20.2/30 interface=ether2
[admin@R01.NY] /ip address> /routing ospf instance set default router-id=10.0.0.20
[admin@R01.NY] /ip address> /routing ospf network
[admin@R01.NY] /routing ospf network> add network=10.0.0.20/32 area=backbone
[admin@R01.NY] /routing ospf network> add network=10.21.20.0/30 area=backbone
[admin@R01.NY] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.20 lsr-id=10.0.0.20
[admin@R01.NY] /routing ospf network> /mpls ldp interface add interface=ether2
[admin@R01.NY] /routing ospf network> /routing bgp instance set default as=65000 router-id=10.0.0.20
[admin@R01.NY] /routing ospf network> /routing bgp peer add name=to_RR_LND remote-address=10.0.0.21 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.NY] /routing ospf network> /ip route vrf add routing-mark-VRF_DEVOPS interfaces=ether3 route-distinguisher=65000:1 import-route-targets=65000:1 export-route-targets=65000:1
[admin@R01.NY] /routing ospf network> /ip address add address=192.168.20.1/24 interface=ether3
[admin@R01.NY] /routing ospf network> /routing bgp instance vrf add routing-mark-VRF_DEVOPS instance=default redistribute-connected=yes
[admin@R01.NY] /routing ospf network> █
```

Рисунок 6 – NY

На рисунке 7 представлен конфиг для роутера LBN.

```
[admin@R01.LBN] > /interface bridge add name=lo0
[admin@R01.LBN] > /ip address
[admin@R01.LBN] /ip address> add address=10.0.0.31/32 interface=lo0
[admin@R01.LBN] /ip address> add address=10.11.31.2/30 interface=ether2
[admin@R01.LBN] /ip address> add address=10.21.31.2/30 interface=ether3
[admin@R01.LBN] /ip address> add address=10.31.30.1/30 interface=ether4
[admin@R01.LBN] /ip address> /routing ospf instance set default router-id=10.0.0.31
[admin@R01.LBN] /ip address> /routing ospf network
[admin@R01.LBN] /routing ospf network> add network=10.0.0.31/32 area=backbone
[admin@R01.LBN] /routing ospf network> add network=10.11.31.0/30 area=backbone
[admin@R01.LBN] /routing ospf network> add network=10.21.31.0/30 area=backbone
[admin@R01.LBN] /routing ospf network> add network=10.31.30.0/30 area=backbone
[admin@R01.LBN] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.31 lsr-id=10.0.0.31
[admin@R01.LBN] /routing ospf network> /mpls ldp interface
[admin@R01.LBN] /mpls ldp interface> add interface=ether2
[admin@R01.LBN] /mpls ldp interface> add interface=ether3
[admin@R01.LBN] /mpls ldp interface> add interface=ether4
[admin@R01.LBN] /mpls ldp interface> /routing bgp instance set default as=65000 router-id=10.0.0.31 redistribute-connected=yes
[admin@R01.LBN] /mpls ldp interface> /routing bgp peer
[admin@R01.LBN] /routing bgp peer> add name=to_HKI remote-address=10.0.0.11 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.LBN] /routing bgp peer> add name=to_LND remote-address=10.0.0.21 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.LBN] /routing bgp peer> add name=to_SVL remote-address=10.0.0.30 remote-as=65000 update-source=lo0 route-reflect=yes address-families=vpn4
[admin@R01.LBN] /routing bgp peer> █
```

Рисунок 7 – LBN

На рисунке 8 представлен конфиг для роутера SVL.

```
[admin@R01.SVL] > /interface bridge add name=lo0
[admin@R01.SVL] > /ip address
[admin@R01.SVL] /ip address> add address=10.0.0.30/32 interface=lo0
[admin@R01.SVL] /ip address> add address=10.31.30.2/30 interface=ether2
[admin@R01.SVL] /ip address> /routing ospf instance set default router-id=10.0.0.30
[admin@R01.SVL] /ip address> /routing ospf network
[admin@R01.SVL] /routing ospf network> add network=10.0.0.30/32 area=backbone
[admin@R01.SVL] /routing ospf network> add network=10.31.30.0/30 area=backbone
[admin@R01.SVL] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.30 lsr-id=10.0.0.30
[admin@R01.SVL] /routing ospf network> /mpls ldp interface add interface=ether2
[admin@R01.SVL] /routing ospf network> /routing bgp instance set default as=65000 router-id=10.0.0.30
[admin@R01.SVL] /routing ospf network> /routing bgp peer add name=to_RR_LBN remote-address=10.0.0.31 remote-as=65000 update-source=lo0 address-families=vpn4
[admin@R01.SVL] /routing ospf network> /ip route vrf add routing-mark-VRF_DEVOPS interfaces=ether3 route-distinguisher=65000:1 import-route-targets=65000:1 export-route-targets=65000:1
[admin@R01.SVL] /routing ospf network> /ip address add address=192.168.30.1/24 interface=ether3
[admin@R01.SVL] /routing ospf network> /routing bgp instance vrf add routing-mark-VRF_DEVOPS instance=default redistribute-connected=yes
[admin@R01.SVL] /routing ospf network> █
```

Рисунок 8 – SVL

На рисунке 9 представлена проверка связности и пингов с PC1.

```
root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4# docker exec -it clab-lab4-PC1 sh
/ # ping 192.168.20.253
PING 192.168.20.253 (192.168.20.253): 56 data bytes
64 bytes from 192.168.20.253: seq=0 ttl=60 time=4.053 ms
64 bytes from 192.168.20.253: seq=1 ttl=60 time=4.188 ms
64 bytes from 192.168.20.253: seq=2 ttl=60 time=3.354 ms
^C
--- 192.168.20.253 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 3.354/3.865/4.188 ms
/ # traceroute 192.168.20.253
traceroute to 192.168.20.253 (192.168.20.253), 30 hops max, 46 byte packets
 1 * * *
 2 * * *
 3 * * *
 4 * * *
 5 192.168.20.253 (192.168.20.253) 3.099 ms 2.780 ms 2.729 ms
/ #
root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4# docker exec -it clab-lab4-PC2 sh
/ # ^C

/ #
root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4# docker exec -it clab-lab4-PC1 sh
/ # ping 192.168.30.253
PING 192.168.30.253 (192.168.30.253): 56 data bytes
64 bytes from 192.168.30.253: seq=0 ttl=60 time=3.581 ms
64 bytes from 192.168.30.253: seq=1 ttl=60 time=3.113 ms
64 bytes from 192.168.30.253: seq=2 ttl=60 time=3.633 ms
^C
--- 192.168.30.253 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 3.113/3.442/3.633 ms
/ # traceroute 192.168.30.253
traceroute to 192.168.30.253 (192.168.30.253), 30 hops max, 46 byte packets
 1 * * *
 2 * * *
 3 * * *
 4 * * *
 5 192.168.30.253 (192.168.30.253) 3.605 ms 2.977 ms 2.647 ms
/ # █
```

Рисунок 9 – PC1

На рисунке 10 представлена проверка связности и пингов с PC2.

```

root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4# docker exec -it clab-lab4-PC2 sh
/ # ping 192.168.10.253
PING 192.168.10.253 (192.168.10.253): 56 data bytes
64 bytes from 192.168.10.253: seq=0 ttl=60 time=2.987 ms
64 bytes from 192.168.10.253: seq=1 ttl=60 time=3.196 ms
64 bytes from 192.168.10.253: seq=2 ttl=60 time=3.693 ms
^C
--- 192.168.10.253 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 2.987/3.292/3.693 ms
/ # traceroute 192.168.10.253
traceroute to 192.168.10.253 (192.168.10.253), 30 hops max, 46 byte packets
 1 * * *
 2 * * *
 3 * * *
 4 * * *
 5 192.168.10.253 (192.168.10.253) 3.128 ms 2.710 ms 3.060 ms
/ # ping 192.168.30.253
PING 192.168.30.253 (192.168.30.253): 56 data bytes
64 bytes from 192.168.30.253: seq=0 ttl=60 time=2.952 ms
64 bytes from 192.168.30.253: seq=1 ttl=60 time=3.442 ms
64 bytes from 192.168.30.253: seq=2 ttl=60 time=3.413 ms
^C
--- 192.168.30.253 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 2.952/3.269/3.442 ms
/ # traceroute 192.168.30.253
traceroute to 192.168.30.253 (192.168.30.253), 30 hops max, 46 byte packets
 1 * * *
 2 * * *
 3 * * *
 4 * * *
 5 192.168.30.253 (192.168.30.253) 3.197 ms 3.095 ms 2.607 ms
/ # █

```

Рисунок 10 – PC2

На рисунке 11 представлена проверка связности и пингов с PC3.



```

root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4# docker exec -it clab-lab4-PC3 sh
/ # ping 192.168.10.253
PING 192.168.10.253 (192.168.10.253): 56 data bytes
64 bytes from 192.168.10.253: seq=0 ttl=60 time=2.966 ms
64 bytes from 192.168.10.253: seq=1 ttl=60 time=3.157 ms
64 bytes from 192.168.10.253: seq=2 ttl=60 time=3.529 ms
^C
--- 192.168.10.253 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 2.966/3.217/3.529 ms
/ # traceroute 192.168.10.253
traceroute to 192.168.10.253 (192.168.10.253), 30 hops max, 46 byte packets
 1 * * *
 2 * * *
 3 * * *
 4 * * *
 5 192.168.10.253 (192.168.10.253) 3.619 ms 3.053 ms 2.793 ms
/ # ping 192.168.20.253
PING 192.168.20.253 (192.168.20.253): 56 data bytes
64 bytes from 192.168.20.253: seq=0 ttl=60 time=2.592 ms
64 bytes from 192.168.20.253: seq=1 ttl=60 time=3.000 ms
64 bytes from 192.168.20.253: seq=2 ttl=60 time=3.392 ms
64 bytes from 192.168.20.253: seq=3 ttl=60 time=3.509 ms
64 bytes from 192.168.20.253: seq=4 ttl=60 time=3.077 ms
64 bytes from 192.168.20.253: seq=5 ttl=60 time=3.655 ms
64 bytes from 192.168.20.253: seq=6 ttl=60 time=3.640 ms
64 bytes from 192.168.20.253: seq=7 ttl=60 time=3.281 ms
64 bytes from 192.168.20.253: seq=8 ttl=60 time=2.920 ms
64 bytes from 192.168.20.253: seq=9 ttl=60 time=3.131 ms
64 bytes from 192.168.20.253: seq=10 ttl=60 time=3.298 ms
^C
--- 192.168.20.253 ping statistics ---
11 packets transmitted, 11 packets received, 0% packet loss
round-trip min/avg/max = 2.592/3.226/3.655 ms
/ # traceroute 192.168.20.253
traceroute to 192.168.20.253 (192.168.20.253), 30 hops max, 46 byte packets
 1 * * *
 2 * * *
 3 * * *
 4 * * *
 5 192.168.20.253 (192.168.20.253) 3.244 ms 3.289 ms 2.921 ms
/ # █

```

Рисунок 11 – PC3

## 2 ЧАСТЬ 2

Во второй части было необходимо: разобрать VRF на 3 роутерах (или отвязать их от интерфейсов), настроить VPLS на 3 роутерах, настроить IP адресацию на PC1,2,3 в одной сети, проверить связность.

На рисунках 12 и 13 представлен уaml-файл для второй части работы.

```

name: lab4_2

mgmt:
  network: mgmt
  ipv4-subnet: 172.20.20.0/24

topology:
  defaults:
    kind: vr-ros
    image: vrnetlab/mikrotik_routeros:6.47.9
  nodes:
    R01.HKI:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.10
    R01.SPB:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.11
    PC1:
      kind: linux
      image: alpine:latest
      cmd: sleep infinity
      network-mode: none
      exec:
        - ip addr add 192.168.0.1/24 dev eth1
    R01.LND:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.12
    R01.NY:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.13
    PC2:
      kind: linux
      image: alpine:latest
      cmd: sleep infinity
      network-mode: none
      exec:
        - ip addr add 192.168.0.2/24 dev eth1
    R01.LBN:
      kind: vr-ros
      mgmt-ipv4: 172.20.20.14

```

Рисунок 12 – yaml

```

PC3:
  kind: linux
  image: alpine:latest
  cmd: sleep infinity
  network-mode: none
  exec:
    - ip addr add 192.168.0.3/24 dev eth1
links:
  - endpoints: ["R01.HKI:eth1", "R01.LND:eth1"]
  - endpoints: ["R01.HKI:eth2", "R01.LBN:eth1"]
  - endpoints: ["R01.LND:eth2", "R01.LBN:eth2"]
  - endpoints: ["R01.HKI:eth3", "R01.SPB:eth1"]
  - endpoints: ["R01.LND:eth3", "R01.NY:eth1"]
  - endpoints: ["R01.LBN:eth3", "R01.SVL:eth1"]
  - endpoints: ["R01.SPB:eth2", "PC1:eth1"]
  - endpoints: ["R01.NY:eth2", "PC2:eth1"]
  - endpoints: ["R01.SVL:eth2", "PC3:eth1"]

```

Рисунок 13 – yaml

На рисунке 14 представлен конфиг для роутера HKI.

```
[admin@R01.HKI] > /interface bridge add name=lo0
[admin@R01.HKI] > /ip address
[admin@R01.HKI] /ip address> add address=10.0.0.11/32 interface=lo0
[admin@R01.HKI] /ip address> add address=10.11.21.1/30 interface=ether2
[admin@R01.HKI] /ip address> add address=10.11.31.1/30 interface=ether3
[admin@R01.HKI] /ip address> add address=10.11.10.1/30 interface=ether4
[admin@R01.HKI] /ip address> /routing ospf instance set default router-id=10.0.0.11
[admin@R01.HKI] /ip address> /routing ospf network
[admin@R01.HKI] /routing ospf network> add network=10.0.0.11/32 area=backbone
[admin@R01.HKI] /routing ospf network> add network=10.11.21.0/30 area=backbone
[admin@R01.HKI] /routing ospf network> add network=10.11.31.0/30 area=backbone
[admin@R01.HKI] /routing ospf network> add network=10.11.10.0/30 area=backbone
[admin@R01.HKI] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.11 lsr-id=10.0.0.11
[admin@R01.HKI] /routing ospf network> /mpls ldp interface
[admin@R01.HKI] /mpls ldp interface> add interface=ether2
[admin@R01.HKI] /mpls ldp interface> add interface=ether3
[admin@R01.HKI] /mpls ldp interface> add interface=ether4
[admin@R01.HKI] /mpls ldp interface> /routing bgp instance set default as=65000 router-id=10.0.0.11 redistribute-connected=yes client-to-client-reflection=yes
[admin@R01.HKI] /routing bgp peer> /routing bgp peer
[admin@R01.HKI] /routing bgp peer> add name=to_LND remote-address=10.0.0.21 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.HKI] /routing bgp peer> add name=to_LBN remote-address=10.0.0.31 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.HKI] /routing bgp peer> add name=to_SPB remote-address=10.0.0.10 remote-as=65000 update-source=lo0 route-reflect=yes address-families=vpn4,l2vpn
[admin@R01.HKI] /routing bgp peer> [
```

Рисунок 14 – HKI

На рисунке 15 представлен конфиг для роутера SPB.

```
[admin@R01.SPB] > /interface bridge add name=lo0
[admin@R01.SPB] > /ip address
[admin@R01.SPB] /ip address> add address=10.0.0.10/32 interface=lo0
[admin@R01.SPB] /ip address> add address=10.11.10.2/30 interface=ether2
[admin@R01.SPB] /ip address> /routing ospf instance set default router-id=10.0.0.10
[admin@R01.SPB] /ip address> /routing ospf network
[admin@R01.SPB] /routing ospf network> add network=10.0.0.10/32 area=backbone
[admin@R01.SPB] /routing ospf network> add network=10.11.10.0/30 area=backbone
[admin@R01.SPB] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.10 lsr-id=10.0.0.10
[admin@R01.SPB] /routing ospf network> /mpls ldp interface add interface=ether2
[admin@R01.SPB] /routing ospf network> /routing bgp peer add name=to_RR_HKI remote-address=10.0.0.11 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.SPB] /routing ospf network> /interface bridge add name=br-vpls
[admin@R01.SPB] /routing ospf network> /interface bridge port add bridge=br-vpls interface=ether3
[admin@R01.SPB] /routing ospf network> /interface vpls bgp-vpls add name=vpls_bgp bridge=br-vpls site-id=10 route-distinguisher=65000:100 export-route-targets=65000:100 import-route-targets=65000:100
[admin@R01.SPB] /routing ospf network> [
```

Рисунок 15 – SPB

На рисунке 16 представлен конфиг для роутера LND.

```
[admin@R01.LND] > /interface bridge add name=lo0
[admin@R01.LND] > /ip address
[admin@R01.LND] /ip address> add address=10.0.0.21/32 interface=lo0
[admin@R01.LND] /ip address> add address=10.11.21.2/30 interface=ether2
[admin@R01.LND] /ip address> add address=10.21.31.1/30 interface=ether3
[admin@R01.LND] /ip address> add address=10.21.20.1/30 interface=ether4
[admin@R01.LND] /ip address> /routing ospf instance set default router-id=10.0.0.21
[admin@R01.LND] /ip address> /routing ospf network
[admin@R01.LND] /routing ospf network> add network=10.0.0.21/32 area=backbone
[admin@R01.LND] /routing ospf network> add network=10.11.21.0/30 area=backbone
[admin@R01.LND] /routing ospf network> add network=10.21.31.0/30 area=backbone
[admin@R01.LND] /routing ospf network> add network=10.21.20.0/30 area=backbone
[admin@R01.LND] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.21 lsr-id=10.0.0.21
[admin@R01.LND] /routing ospf network> /mpls ldp interface
[admin@R01.LND] /mpls ldp interface> add interface=ether2
[admin@R01.LND] /mpls ldp interface> add interface=ether3
[admin@R01.LND] /mpls ldp interface> add interface=ether4
[admin@R01.LND] /mpls ldp interface> /routing bgp instance set default as=65000 router-id=10.0.0.21 redistribute-connected=yes client-to-client-reflection=yes
[admin@R01.LND] /routing bgp peer> /routing bgp peer
[admin@R01.LND] /routing bgp peer> add name=to_HKI remote-address=10.0.0.11 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.LND] /routing bgp peer> add name=to_LBN remote-address=10.0.0.31 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.LND] /routing bgp peer> add name=to_NYD remote-address=10.0.0.20 remote-as=65000 update-source=lo0 route-reflect=yes address-families=vpn4,l2vpn
[admin@R01.LND] /routing bgp peer> [
```

Рисунок 16 – LND

На рисунке 17 представлен конфиг для роутера NY.

```
[admin@R01.NY] > /interface bridge add name=lo0
[admin@R01.NY] > /ip address
[admin@R01.NY] /ip address> add address=10.0.0.20/32 interface=lo0
[admin@R01.NY] /ip address> add address=10.21.20.2/30 interface=ether2
[admin@R01.NY] /ip address> /routing ospf instance set default router-id=10.0.0.20
[admin@R01.NY] /ip address> /routing ospf network
[admin@R01.NY] /routing ospf network> add network=10.0.0.20/32 area=backbone
[admin@R01.NY] /routing ospf network> add network=10.21.20.0/30 area=backbone
[admin@R01.NY] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.20 lsr-id=10.0.0.20
[admin@R01.NY] /routing ospf network> /mpls ldp interface add interface=ether2
[admin@R01.NY] /routing ospf network> /routing bgp peer add name=to_RR_LND remote-address=10.0.0.21 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.NY] /routing ospf network> /interface bridge add name=br-vpls
[admin@R01.NY] /routing ospf network> /interface bridge port add bridge=br-vpls interface=ether3
[admin@R01.NY] /routing ospf network> /interface vpls bgp-vpls add name=vpls_bgp bridge=br-vpls site-id=20 route-distinguisher=65000:100 export-route-targets=65000:100 import-route-targets=65000:100
[admin@R01.NY] /routing ospf network> [
```

Рисунок 17 – NY

На рисунке 18 представлен конфиг для роутера LBN.

```
[admin@R01.LBN] > /interface bridge add name=lo0
[admin@R01.LBN] > /ip address
[admin@R01.LBN] /ip address> add address=10.0.0.31/32 interface=lo0
[admin@R01.LBN] /ip address> add address=10.11.31.2/30 interface=ether2
[admin@R01.LBN] /ip address> add address=10.21.31.2/30 interface=ether3
[admin@R01.LBN] /ip address> add address=10.31.30.1/30 interface=ether4
[admin@R01.LBN] /ip address> /routing ospf instance set default router-id=10.0.0.31
[admin@R01.LBN] /ip address> /routing ospf network
[admin@R01.LBN] /routing ospf network> add network=10.0.0.31/32 area=backbone
[admin@R01.LBN] /routing ospf network> add network=10.11.31.0/30 area=backbone
[admin@R01.LBN] /routing ospf network> add network=10.21.31.0/30 area=backbone
[admin@R01.LBN] /routing ospf network> add network=10.31.30.0/30 area=backbone
[admin@R01.LBN] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.31 lsr-id=10.0.0.31
[admin@R01.LBN] /routing ospf network> /mpls ldp interface
[admin@R01.LBN] /mpls ldp interface> add interface=ether2
[admin@R01.LBN] /mpls ldp interface> add interface=ether3
[admin@R01.LBN] /mpls ldp interface> add interface=ether4
[admin@R01.LBN] /mpls ldp interface> /routing bgp instance set default as=65000 router-id=10.0.0.31 redistribute-connected=yes client-to-client-reflection=yes
[admin@R01.LBN] /routing bgp peer> add name=to_HKI remote-address=10.0.0.11 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.LBN] /routing bgp peer> add name=to_LND remote-address=10.0.0.21 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.LBN] /routing bgp peer> add name=to_SVL remote-address=10.0.0.30 remote-as=65000 update-source=lo0 route-reflect=yes address-families=vpn4,l2vpn
[admin@R01.LBN] /routing bgp peer> [
```

Рисунок 18 – LBN

На рисунке 19 представлен конфиг для роутера SVL.

```
[admin@R01.SVL] > /interface bridge add name=lo0
[admin@R01.SVL] > /ip address
[admin@R01.SVL] /ip address> add address=10.0.0.30/32 interface=lo0
[admin@R01.SVL] /ip address> add address=10.31.30.2/30 interface=ether2
[admin@R01.SVL] /ip address> /routing ospf instance set default router-id=10.0.0.30
[admin@R01.SVL] /ip address> /routing ospf network
[admin@R01.SVL] /routing ospf network> add network=10.0.0.30/32 area=backbone
[admin@R01.SVL] /routing ospf network> add network=10.31.30.0/30 area=backbone
[admin@R01.SVL] /routing ospf network> /mpls ldp set enabled=yes transport-address=10.0.0.30 lsr-id=10.0.0.30
[admin@R01.SVL] /routing ospf network> /mpls ldp interface add interface=ether2
[admin@R01.SVL] /routing ospf network> /routing bgp instance set default as=65000 router-id=10.0.0.30
[admin@R01.SVL] /routing ospf network> /routing bgp peer add name=to_R01.LBN remote-address=10.0.0.31 remote-as=65000 update-source=lo0 address-families=vpn4,l2vpn
[admin@R01.SVL] /routing ospf network> /interface bridge port add bridge=br-vpls interface=ether3
[admin@R01.SVL] /routing ospf network> /interface vpls bgp-vpls add name=vpls_bgp bridge=br-vpls site-id=30 route-distinguisher=65000:100 export-route-targets=65000:100 import-route-targets=65000:100
[admin@R01.SVL] /routing ospf network> [
```

Рисунок 19 – SVL

На рисунке 20 представлена проверка связности и пингов с PC1.

```
root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4_2# docker exec -it clab-lab4_2-PC1 sh
/ # ping 192.168.0.2
PING 192.168.0.2 (192.168.0.2): 56 data bytes
64 bytes from 192.168.0.2: seq=0 ttl=64 time=1.182 ms
64 bytes from 192.168.0.2: seq=1 ttl=64 time=2.298 ms
64 bytes from 192.168.0.2: seq=2 ttl=64 time=2.141 ms
^C
--- 192.168.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.182/1.873/2.298 ms
/ # traceroute 192.168.0.2
traceroute to 192.168.0.2 (192.168.0.2), 30 hops max, 46 byte packets
 1 192.168.0.2 (192.168.0.2) 1.328 ms 1.658 ms 0.856 ms
/ #
root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4_2# docker exec -it clab-lab4_2-PC2 sh
/ #
root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4_2# docker exec -it clab-lab4_2-PC1 sh
/ # ping 192.168.0.3
PING 192.168.0.3 (192.168.0.3): 56 data bytes
64 bytes from 192.168.0.3: seq=0 ttl=64 time=1.146 ms
64 bytes from 192.168.0.3: seq=1 ttl=64 time=1.308 ms
64 bytes from 192.168.0.3: seq=2 ttl=64 time=1.294 ms
^C
--- 192.168.0.3 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.146/1.249/1.308 ms
/ # traceroute 192.168.0.3
traceroute to 192.168.0.3 (192.168.0.3), 30 hops max, 46 byte packets
 1 192.168.0.3 (192.168.0.3) 0.702 ms 0.724 ms 0.425 ms
^C
```

Рисунок 20 – PC1

На рисунке 21 представлена проверка связности и пингов с PC2.

```
root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4_2# docker exec -it clab-lab4_2-PC2 sh
/ # ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
64 bytes from 192.168.0.1: seq=0 ttl=64 time=1.213 ms
64 bytes from 192.168.0.1: seq=1 ttl=64 time=2.109 ms
64 bytes from 192.168.0.1: seq=2 ttl=64 time=2.308 ms
^C
--- 192.168.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.213/1.876/2.308 ms
/ # traceroute 192.168.0.1
traceroute to 192.168.0.1 (192.168.0.1), 30 hops max, 46 byte packets
 1 192.168.0.1 (192.168.0.1) 1.275 ms 1.788 ms 1.020 ms
/ # ping 192.168.0.3
PING 192.168.0.3 (192.168.0.3): 56 data bytes
64 bytes from 192.168.0.3: seq=0 ttl=64 time=1.299 ms
64 bytes from 192.168.0.3: seq=1 ttl=64 time=1.355 ms
64 bytes from 192.168.0.3: seq=2 ttl=64 time=1.285 ms
^C
--- 192.168.0.3 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.285/1.313/1.355 ms
/ # traceroute 192.168.0.3
traceroute to 192.168.0.3 (192.168.0.3), 30 hops max, 46 byte packets
 1 192.168.0.3 (192.168.0.3) 0.815 ms 1.517 ms 0.734 ms
/ #
```

Рисунок 21 – PC2

На рисунке 22 представлена проверка связности и пингов с PC3.

```
root@seraypuh:/home/seraypuh/vrnetlab/mikrotik/routeros/lab4_2# docker exec -it clab-lab4_2-PC3 sh
/ # ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
64 bytes from 192.168.0.1: seq=0 ttl=64 time=0.970 ms
64 bytes from 192.168.0.1: seq=1 ttl=64 time=0.849 ms
64 bytes from 192.168.0.1: seq=2 ttl=64 time=1.661 ms
^C
--- 192.168.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.849/1.160/1.661 ms
/ # traceroute 192.168.0.1
traceroute to 192.168.0.1 (192.168.0.1), 30 hops max, 46 byte packets
 1 192.168.0.1 (192.168.0.1) 0.730 ms 1.417 ms 0.606 ms
/ # ping 192.168.0.2
PING 192.168.0.2 (192.168.0.2): 56 data bytes
64 bytes from 192.168.0.2: seq=0 ttl=64 time=0.753 ms
64 bytes from 192.168.0.2: seq=1 ttl=64 time=1.754 ms
64 bytes from 192.168.0.2: seq=2 ttl=64 time=1.615 ms
^C
--- 192.168.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.753/1.374/1.754 ms
/ # traceroute 192.168.0.1
traceroute to 192.168.0.1 (192.168.0.1), 30 hops max, 46 byte packets
 1 ^C
/ # traceroute 192.168.0.2
traceroute to 192.168.0.2 (192.168.0.2), 30 hops max, 46 byte packets
 1 192.168.0.2 (192.168.0.2) 0.732 ms 1.377 ms 0.535 ms
/ #
```

Рисунок 22 – PC3



### 3 СХЕМА СЕТИ

На рисунке 23 представлена схема сети.

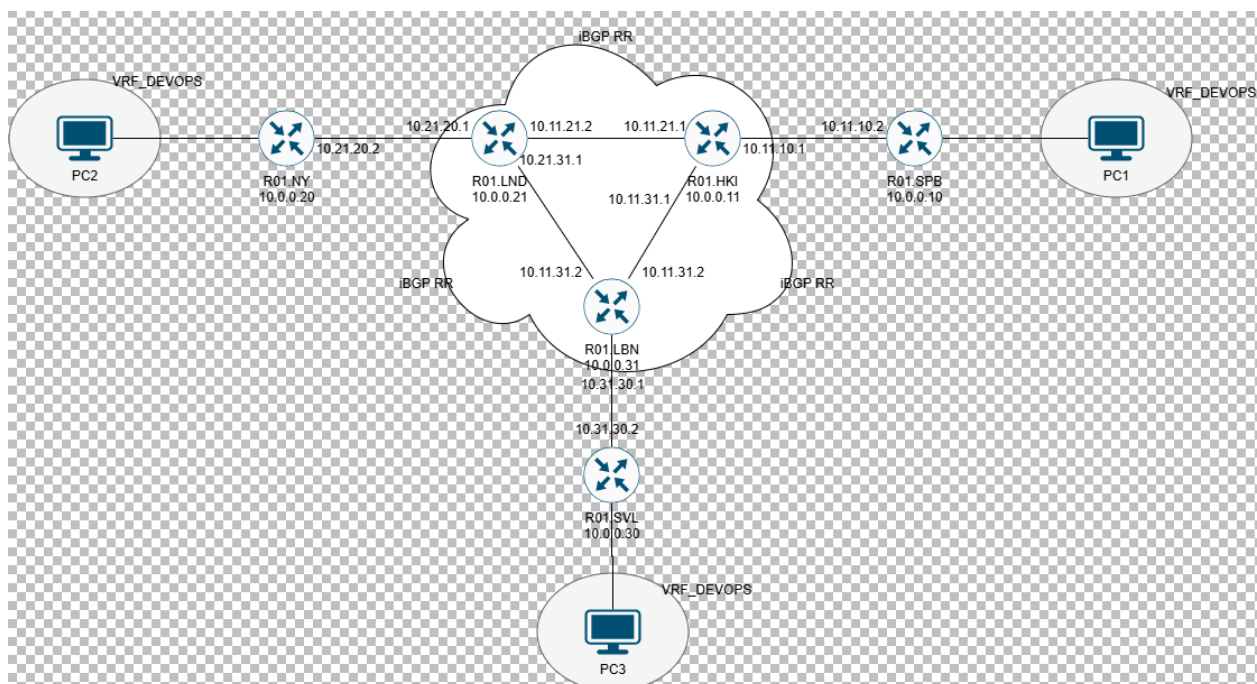


Рисунок 23 – Схема сети

## **ЗАКЛЮЧЕНИЕ**

В ходе данной лабораторной работы были получены навыки работы с протоколами BGP, MPLS, L3VPN и MPLS.